

Tissue- and phenotype-dependent transposable-element responses to ocean acidification

Exploratory brain–liver RNA-seq reanalysis in the coral reef fish *Acanthochromis polyacanthus*

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EXPLORATORY PILOT

Analysis freeze: 30 July 2026

60 samples • 30 brain • 30 liver

QUESTION

Do developmental and intergenerational CO₂ exposures produce tissue- and parental-phenotype-specific TE responses?

STUDY DESIGN

n = 60
2 tissues × 3 exposure groups × 2 parental phenotypes
5 biological samples per cell

D–C developmental vs control
I–C intergenerational vs control
I–D intergenerational vs developmental

ANALYSIS

- Species-specific de novo TE annotation
- STAR alignment retaining multimappers
- TEcount multi-mode quantification
- Family-aware DESeq2 model: ~ family + treatment
- Adjusted TE–gene and GABA-pathway screens



VALIDATION STATUS

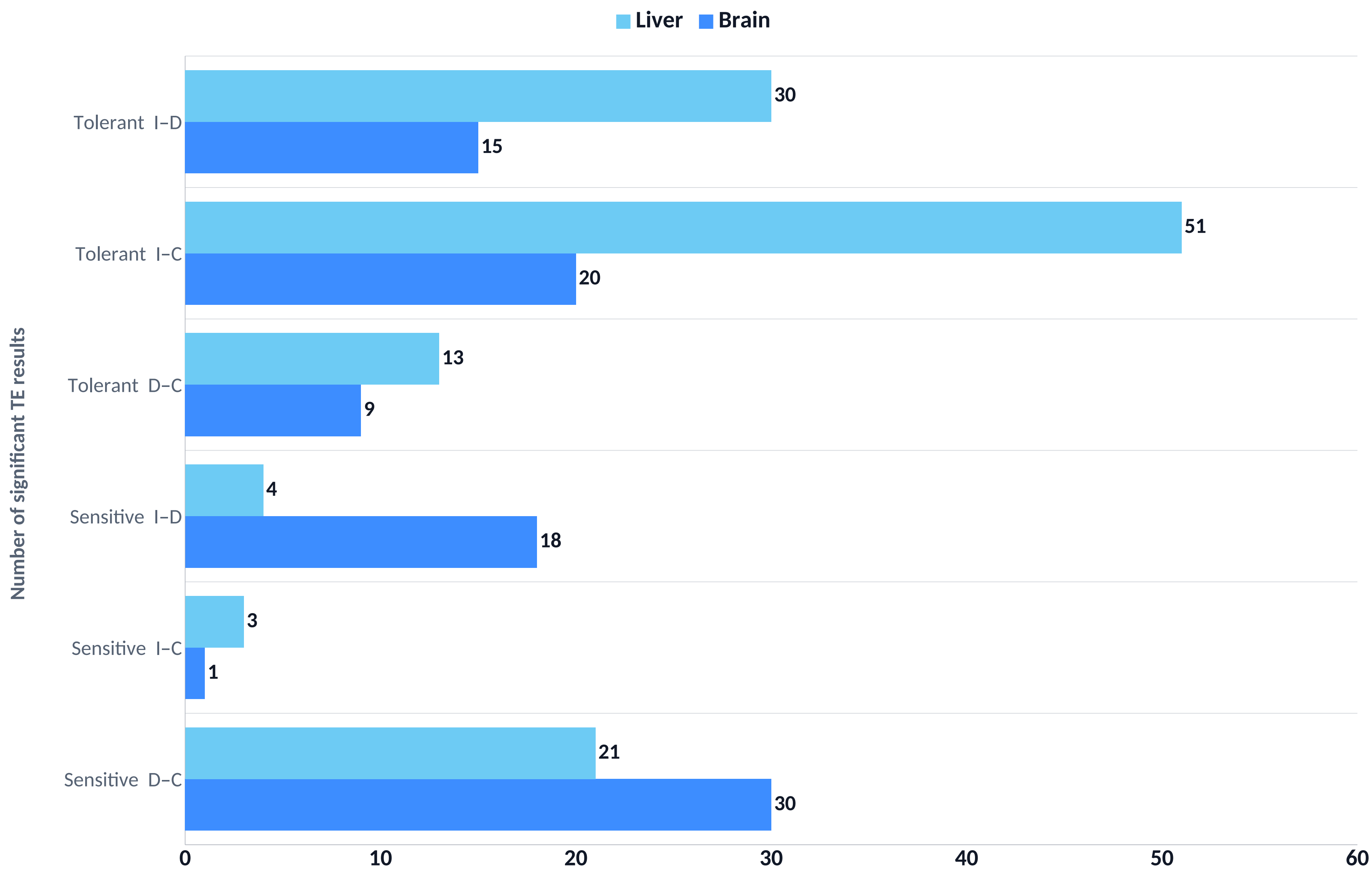
A 162-library brain–gill–liver cohort is frozen for full validation (54 libraries per tissue; 9 per experimental cell). Results shown here are the completed pilot only.

PRIMARY RESULT

The dominant TE response shifts from sensitive-parent brain after developmental exposure to tolerant-parent liver after intergenerational exposure.

Significant strict-TE results by contrast

FDR < 0.05 • n = 5 per tissue × exposure × phenotype cell



30

Sensitive D–C
brain

contrasts with

51

Tolerant I–C
liver

73 brain
93 liver

unique significant strict-TE features across tested strata

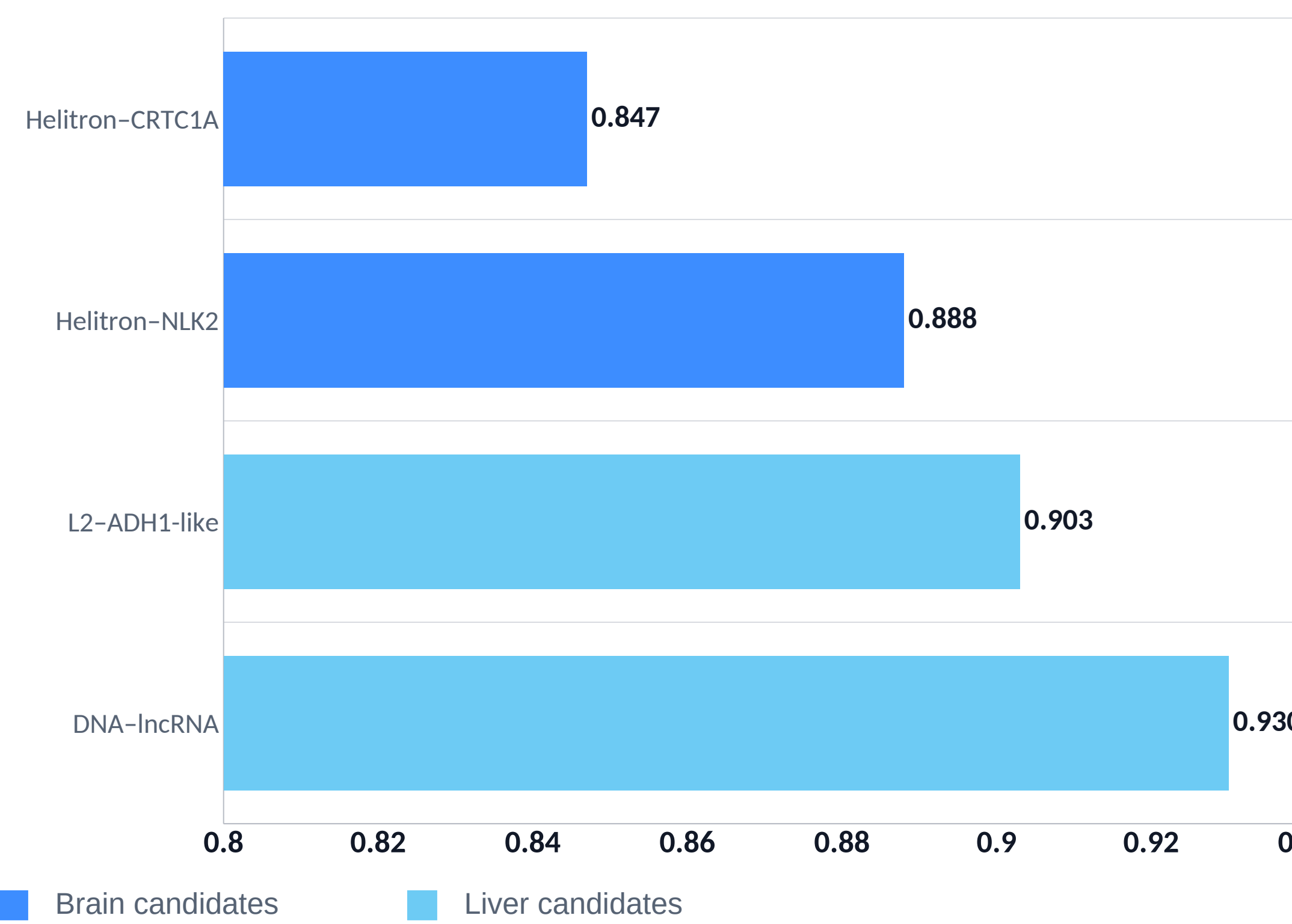
INTERPRETATION

Parental phenotype modifies both the magnitude and tissue localization of TE responses. The pattern is consistent with distinct developmental and intergenerational regulatory states, but requires confirmation in the full cohort.

CANDIDATE ASSOCIATIONS

Top TE–gene relationships remained strong after adjustment for family, treatment and phenotype.

Residual Spearman ρ • n = 30 per tissue



NEURAL HYPOTHESIS

3

direct GABA
genes
in tolerant brain I
–C

GABA transporter 1-like
gabra5
gabrb3

gabra5 and gabrb3 were also significant for I–D. These signals motivate targeted tests of inhibitory-neural regulation.

WHAT THIS SUPPORTS

- Tissue identity is central to TE-response interpretation.
- Parental phenotype changes which exposure contrast dominates.
- TE–gene correlations prioritize mechanistic candidates; they do not establish regulation.

NEXT STEP

Validate effect direction, tissue specificity and GABA-linked candidates in the frozen 162-library cohort.